Adaptive Metaheuristic Optimization of Periodic Autocorrelation in Binary Sequences for Next-Generation GNSS

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***Abstract*—Binary sequences exhibiting low Peak Sidelobe Levels (PSL) in their periodic autocorrelation function are paramount for accurate ranging in Global Navigation Satellite Systems (GNSS). While classical algebraic constructions (Legendre, Weil, Sidelnikov) provide bounded properties, their PSL scales sub-optimally. Existing heuristic optimizations suffer from $O(N \log N)$ computational complexity per bit-flip or fall into local optima due to spectral quantization noise and destructive crossover. This paper proposes a rigorous comparative analysis of four advanced frameworks: Genetic Algorithm with Local Search (GA-LS), Hybrid Simulated Annealing (HSA), PeCAN integrated with Large Neighborhood Search (PeCAN-LNS), and the Brest-Boskovic (BB) metaheuristic. By uniformly applying a novel $O(N)$ FFT-based delta-update mechanism across all algorithms, we evaluate their efficacy on lengths up to 10230 (exact GPS L5/BeiDou B1C length). Extensive simulations demonstrate that the BB algorithm, utilizing an adaptive L_γ-norm cost function, definitively outperforms evolutionary and spectral approaches.**

***Index Terms*—Binary Sequences, Peak Sidelobe Level (PSL), GNSS, Brest-Boskovic Algorithm, Metaheuristics, Autocorrelation.**

I. Introduction

A. Problem Statement

The periodic autocorrelation function (PACF) is a fundamental metric in digital sequence design. For a binary sequence $\mathbf{x} = [x_0, x_1, \ldots, x_{N-1}]$ where $x_i \in \{-1, +1\}$, an ideal PACF exhibits a sharp peak at zero shift and absolute zero energy at all non-zero shifts τ. The deviation from this ideality is quantified by the Peak Sidelobe Level (PSL). High PSL values directly cause false alarms in radar, increase bit-error rates in communications, and induce severe tracking errors in positioning systems.

B. Existing Systems and Research Gap

Deterministic algebraic constructions—namely Legendre, Weil, and Sidelnikov sequences—are heavily relied upon due to their predictable generation. However, their PSL scales at $O(\sqrt{N})$, which is unacceptable for high-precision applications. To mitigate this, heuristic optimizations like Genetic Algorithms (GA) and Simulated Annealing (SA) have been explored. The primary gap in existing literature is twofold:

1) *Computational Inefficiency:* Most heuristic approaches recalculate the full $O(N \log N)$ FFT to evaluate PSL after every single bit-flip, rendering optimization for long sequences practically impossible.
2) *Sub-optimal Search Mechanisms:* Standard GAs destroy beneficial sequence phasing through blind crossover. Conversely, spectral methods like PeCAN suffer severely from quantization noise when projecting continuous spectral domains back to discrete binary values, leading to stagnation at high lengths.

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C. Proposed Work and Methodologies

In this paper, we bridge this gap by evaluating four distinct optimization paradigms unified by a highly efficient $O(N)$ delta-update mathematical formulation. We benchmark: (1) GA-LS, which preserves autocorrelation symmetry via sequence rotation; (2) HSA, which uses multi-bit flips and probabilistic acceptance; (3) PeCAN-LNS, operating in the frequency domain with an LNS "kick" to resolve quantization errors; and (4) the Brest-Boskovic (BB) metaheuristic, which replaces standard PSL objectives with an adaptive L_γ-norm. We mathematically prove that BB overcomes the stagnation issues of existing AI methods, achieving unprecedented PSL reduction.

II. GNSS Sequence Context and Relevance

Global Navigation Satellite Systems (GNSS), such as GPS, Galileo, and BeiDou, rely on Pseudo-Random Noise (PRN) codes for satellite ranging and Code-Division Multiple Access (CDMA). Current modernized GNSS constellations utilize specific prime-length code families; for instance, GPS L5 and BeiDou B1C utilize primary codes of exactly length $N = 10230$, derived from Weil and Legendre sequences.

High periodic PSL in these PRN codes elevates the cross-correlation floor between satellites, exacerbating the "near-far" problem and increasing multipath interference in urban canyons. While current GNSS codes are historically fixed, the push towards Software-Defined GNSS (SD-GNSS) and Inter-Satellite Links (ISL) necessitates dynamically tunable sequences. The algebraic families evaluated in this work (Legendre, Weil, Sidelnikov) are prime candidates for these future flexible architectures. By optimizing their PSL, this work directly translates to narrower tracking jitter loops, improved Signal-to-Noise Ratio (SNR) thresholds, and enhanced anti-jamming resilience.

III. Mathematical Framework

The PACF of sequence $\mathbf{x}$ at shift τ is $R(\tau) = \sum_{i=0}^{N-1} x_i x_{(i+\tau \bmod N)} = \mathrm{Re}\left\{\mathcal{F}^{-1}\left(|X(k)|^2\right)\right\}$. The objective is to minimize $\mathrm{PSL}(\mathbf{x}) = \max_{1 \le \tau \le N/2} |R(\tau)|$.

Algorithm 1 GA-LS Optimization

Require: Seed $\mathbf{x}^{(0)}$, Population size P
Init $P = \{\text{rotate}(\mathbf{x}^{(0)}, i \times \lfloor N/P \rfloor) \mid i \in [0, P-1]\}$
for gen = 1 **to** G_{max} **do**
  Sort P by PSL. Retain top 2 elites.
  while $|P_{new}| < P$ **do**
    Select parents p_1 (elite), p_2 (random top 5)
    Crossover at c: child $= [p_1[0:c], p_2[c:N]]$
    With prob. 0.15, flip random bit $m \in [0, N]$
    Add child to P_{new}
  end while
end for
return Best sequence

Algorithm 2 Hybrid Simulated Annealing

Require: Seed $\mathbf{x}^{(0)}$, Temp T_0, Decay α
$R \leftarrow \text{FFT}(\mathbf{x}^{(0)})$, $\mathbf{x}^* \leftarrow \mathbf{x}^{(0)}$, $T \leftarrow T_0$
for $t = 1$ **to** T_{max} **do**
  Select $k \in \{1, 2, 3\}$ indices. $\mathbf{x}_{new} \leftarrow \text{flip}(\mathbf{x}_{curr}, \text{indices})$
  Compute ΔR via Eq. (2). Calc. PSL_{new}
  $\Delta E \leftarrow PSL_{new} - PSL_{curr}$
  if $\Delta E < 0$ **or** $\text{rand}(0,1) < \exp(-\Delta E/T)$ **then**
    $\mathbf{x}_{curr} \leftarrow \mathbf{x}_{new}$, $R \leftarrow R + \Delta R$
    if $PSL_{new} < PSL^*$ **then**
      $\mathbf{x}^* \leftarrow \mathbf{x}_{curr}$, $PSL^* \leftarrow PSL_{new}$
    end if
  end if
  $T \leftarrow \alpha \cdot T$
end for
return $\mathbf{x}^*$

Algorithm 3 PeCAN-LNS Framework

Require: Seed $\mathbf{x}^{(0)}$, Stagnation limit S_{lim}
$\mathbf{x}_{curr} \leftarrow \mathbf{x}^{(0)}$, stag $\leftarrow 0$
for $t = 1$ **to** T_{max} **do**
  $X \leftarrow \mathcal{F}\{\mathbf{x}_{curr}\}$
  $X_{ideal} \leftarrow (\sqrt{N} + \eta)e^{j\angle X}$, $\eta \sim \mathcal{N}(0, \sigma^2_{stag})$
  $\mathbf{x}_{cont} \leftarrow \text{Re}\{\mathcal{F}^{-1}(X_{ideal})\}$
  $\mathbf{x}_{cand} \leftarrow \text{sgn}(\mathbf{x}_{cont})$
  if $PSL(\mathbf{x}_{cand}) < PSL(\mathbf{x}^*)$ **then**
    $\mathbf{x}^* \leftarrow \mathbf{x}_{cand}$, stag $\leftarrow 0$
  else
    stag $\leftarrow$ stag $+1$
  end if
  if stag $> S_{lim}$ **then**
    Flip indices where $|\mathbf{x}_{cont}| < \text{Percentile}(|\mathbf{x}_{cont}|, 1)$
    stag $\leftarrow 0$
  end if
  $\mathbf{x}_{curr} \leftarrow \mathbf{x}_{cand}$
end for
return $\mathbf{x}^*$

Algorithm 4 Brest-Boskovic Adaptive Opt.

Require: Seed $\mathbf{x}^{(0)}$, $\gamma_0 = 4.0$, $\gamma_{max} = 18.0$
$R \leftarrow \text{FFT}(\mathbf{x}^{(0)})$, $\mathbf{x}^* \leftarrow \mathbf{x}^{(0)}$, $\gamma \leftarrow \gamma_0$
for $t = 1$ **to** T_{max} **do**
  Select random bit index $m \in [0, N]$
  Compute ΔR using Eq. (2)
  $C_{curr} = \sum |R(\tau)|^\gamma$, $C_{new} = \sum |R(\tau) + \Delta R(\tau)|^\gamma$
  if $C_{new} < C_{curr}$ **then**
    Flip x_m. Update $R \leftarrow R + \Delta R$
    if $\max |R(\tau)| < PSL^*$ **then**
      $\mathbf{x}^* \leftarrow \mathbf{x}$, $PSL^* \leftarrow PSL(\mathbf{x})$
      $\gamma \leftarrow \min(\gamma + 0.1, \gamma_{max})$
    end if
  end if
end for
return $\mathbf{x}^*$

To achieve real-time optimization, we avoid recalculating the FFT. If a bit at index m is flipped ($x_m \to -x_m$), the exact variation in the ACF is derived as:

$$\Delta R(\tau) = -2x_m \left[x_{(m-\tau \bmod N)} + x_{(m+\tau \bmod N)}\right] \quad (1)$$

This $O(N)$ vector update is the foundational computational engine for all four proposed algorithms, reducing evaluation time by orders of magnitude compared to standard $O(N \log N)$ FFT recalculations.

IV. Algorithmic Formulations

All algorithms share inputs: Initial Seed $\mathbf{x}^{(0)} \in \{\text{Legendre, Weil, Sidelnikov}\}$, optimally shifted by $N/4$.

V. Results and Discussion

To validate the frameworks, sequences were optimized up to $N = 10230$. GA-LS and HSA were run for 15,000 iterations, PeCAN-LNS for 1,000, and BB for 100,000 (due to its exceptionally lightweight $O(N)$ acceptance logic). Table I presents the comprehensive PSL performance.

The empirical results elucidate critical phenomena regarding the scalability of these algorithms:

1) *The Quantization Ceiling of PeCAN-LNS:* At $N = 10230$, PeCAN-LNS converges to an identical PSL of 154 across Legendre, Weil, and Sidelnikov seeds. This proves that at large lengths, continuous spectral flattening pushes the sequence into a universal binary quantization trap that the LNS kick cannot escape. Conversely, BB retains sequence-specific characteristics, achieving PSLs of 114, 122, and 118.
2) *Crossover Destructiveness in GA-LS:* For $N > 1500$, GA-LS demonstrates the weakest optimization. At $N = 10230$, GA-LS only reduces the Legendre PSL from 326 to 294. Single-point crossover inherently destroys the delicate bipolar phasing required for low autocorrelation, a flaw entirely circumvented by BB's bitwise mutation.
3) *Stochastic Instability of HSA:* HSA shows erratic behavior; at $N = 5115$, it increased the Weil PSL from 213 to 258. The probabilistic acceptance of worse solutions proves highly unstable for the strict combinatorial bounds of binary autocorrelation.
4) *Short-Length Anomalies:* At very short lengths (e.g., $N = 200$ Weil), PeCAN achieves a PSL of 12 compared to BB's 16. At small scales, the global perspective of spectral shaping perfectly matches the binary constraint space. However, this advantage vanishes entirely as N scales beyond 500, establishing BB as the only mathematically scalable solution.

VI. Conclusion and Future Work

This paper presented a rigorous computational comparison of four AI and metaheuristic strategies for minimizing the periodic autocorrelation PSL of binary sequences. By standardizing the evaluation using a mathematically derived $O(N)$ delta-update mechanism, we isolated the algorithmic efficiency of each method. The results conclusively demonstrate that the Brest-Boskovic metaheuristic is the only robust paradigm for large-scale optimization. By leveraging an adaptive L_γ-norm penalty, BB avoids the universal quantization stagnation inherent to PeCAN and the destructive crossover of GAs. By optimizing prime-length algebraic seeds up to exactly 10230, this work establishes a highly efficient pathway for generating

TABLE I
COMPARATIVE PSL PERFORMANCE ACROSS SEQUENCE FAMILIES AND ALL EVALUATED LENGTHS

Length	Seed Type	Initial PSL	GA-LS	HSA	PeCAN-LNS	Brest-Boskovic
100	Legendre	20	12	12	8	**8**
	Weil	16	12	12	8	**8**
	Sidelnikov	4	4	4	4	**4**
200	Legendre	44	28	20	16	**12**
	Weil	28	24	20	**12**	16
	Sidelnikov	24	20	16	**12**	16
1023	Legendre	85	73	67	**35**	**35**
	Weil	77	63	61	41	**33**
	Sidelnikov	49	45	49	41	**33**
1500	Legendre	92	84	88	44	**40**
	Weil	104	92	80	48	**36**
	Sidelnikov	88	80	72	48	**40**
1800	Legendre	124	108	104	56	**44**
	Weil	112	104	88	60	**48**
	Sidelnikov	4	4	4	4	**4**
2046	Legendre	122	110	106	58	**50**
	Weil	118	114	98	62	**46**
	Sidelnikov	102	82	90	62	**46**
5115	Legendre	201	193	187	95	**77**
	Weil	213	205	258	123	**77**
	Sidelnikov	117	117	117	99	**73**
10230	Legendre	326	294	290	154	**114**
	Weil	314	306	177	154	**122**
	Sidelnikov	278	238	246	154	**118**

next-generation PRN codes applicable to advanced GNSS constellations. Future work will focus on optimizing mutual PSL for multi-satellite CDMA constellations and implementing the BB algorithm on FPGAs for real-time Software-Defined GNSS applications.

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